

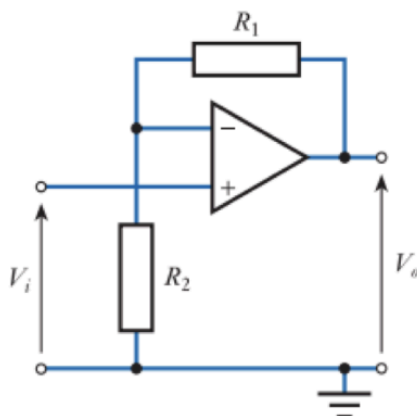
ENGS 152 Circuits Lab

Lab 11

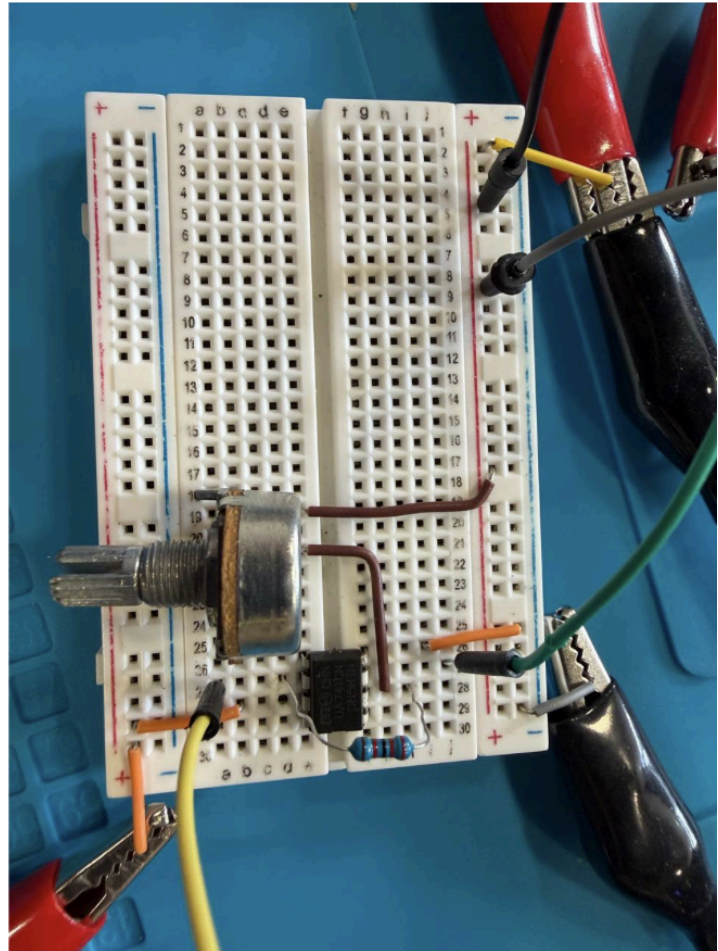
Basic OpAmp Circuits

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Step 2: Non-inverting amplifier



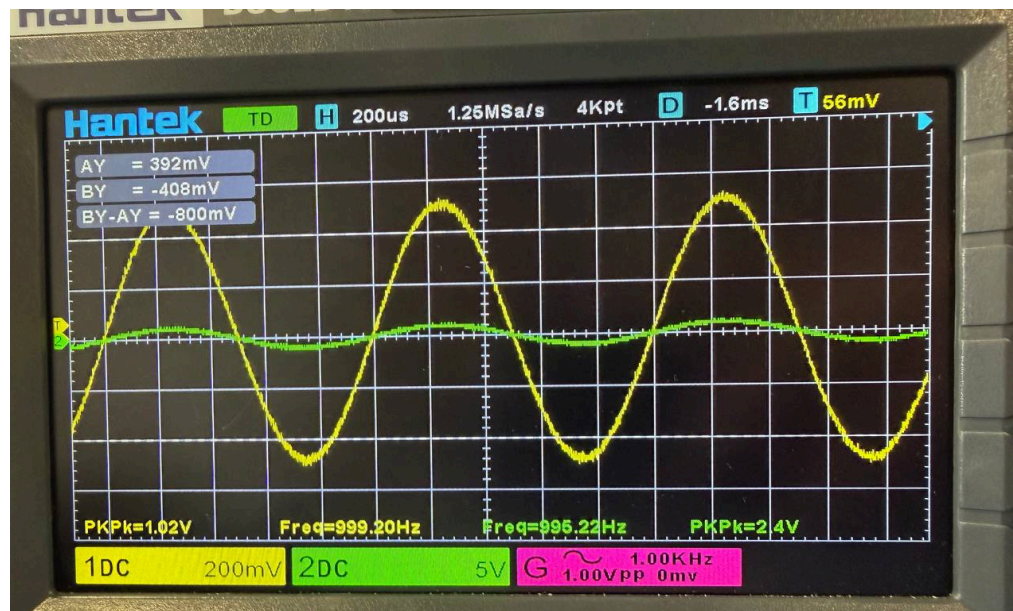
$$G = \frac{V_o}{V_i} = \frac{R_1 + R_2}{R_2}$$



When the potentiometer is rotated fully to one side, our output signal should have been double our input signal, which means the gain is 2. From the equation of the gain and taking into account the available components list, we chose the potentiometer $R_2 = 10k$ and resistor $R_1 = 10k$.

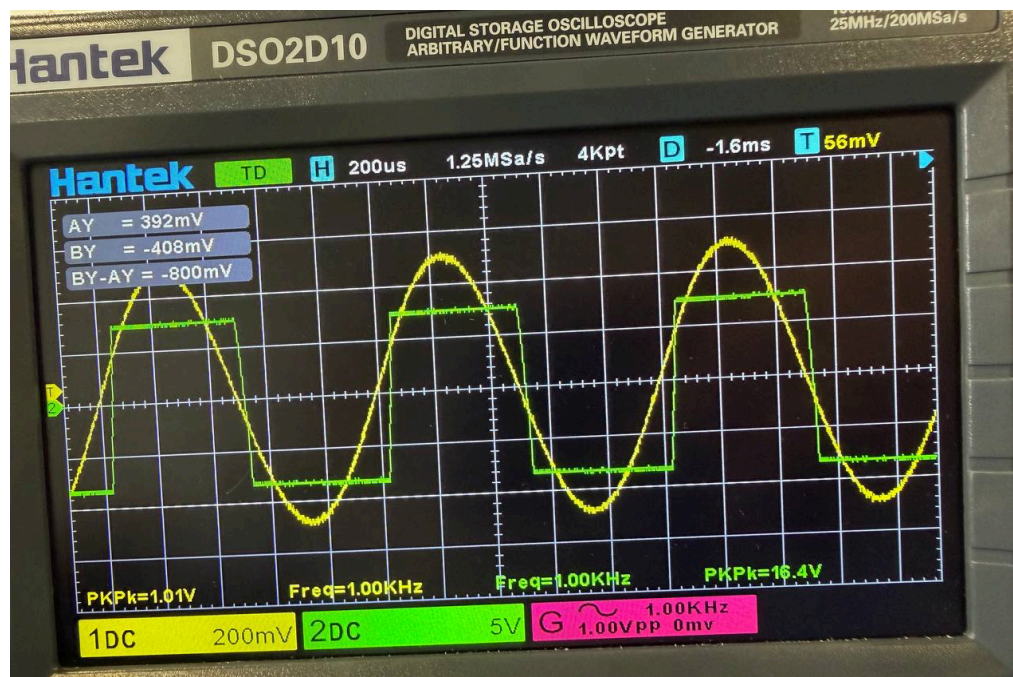
When it is rotated to the other side, our gain should have been large enough to saturate the input signal. On the other side, $R_2 = 0$, which makes the gain very large meaning the output will saturate.

The amplifier's gain is close to the input signal:



Here, if we take a look at our PkPK values we may observe that $V_o/V_i \approx 2$, which is equal to our gain.

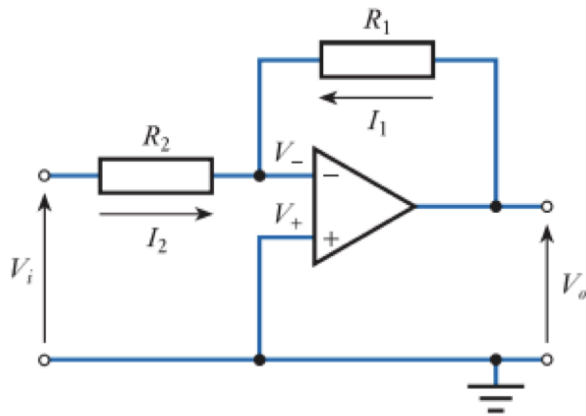
The amplifier is saturated:



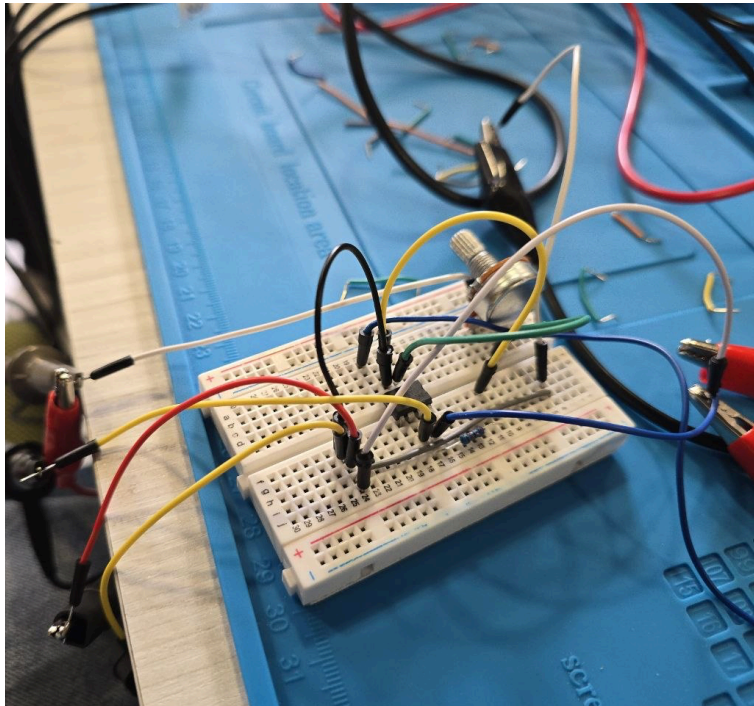
In this picture, the amplifier is fully saturated since the output voltage amplitude surpasses our DC supplies of 9V.

The results we obtained perfectly coincide with the theory.

Step 3: Inverting amplifier

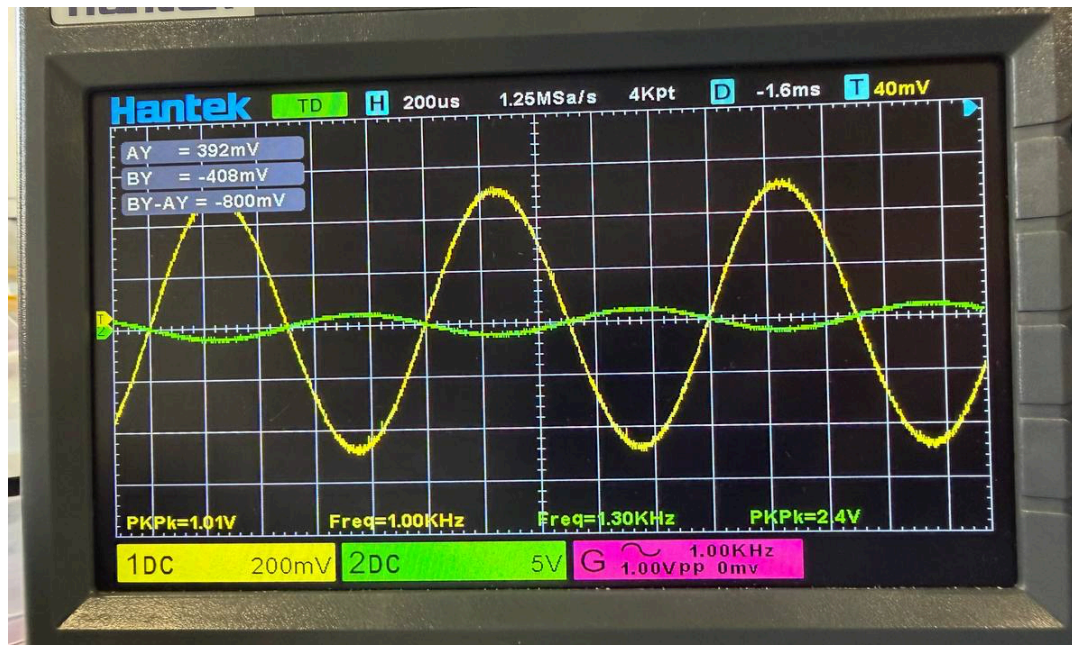


$$G = \frac{V_o}{V_i} = -\frac{R_1}{R_2}$$



When the potentiometer is rotated fully to one side, our output signal should have been inverted and double our input signal, which is why we built an inverting amplifier with gain -2, where potentiometer $R_2 = 10k$ and fixed resistor $R_1 = 22k$ from the available components. When it is rotated to the other side, our gain should be inverted and large enough to

saturate the input signal, meaning the gain should be very large, which is the case when $R_2 = 0$.
The amplifier's gain is close to the input signal:



Here, if we take a look at our PkPk values we may observe that the output signal is inverted and $V_o/V_i \approx 2$, which is equal to our gain.

The amplifier is saturated:



In this picture, the amplifier is fully saturated since the output voltage amplitude surpasses our DC supplies of 9V. The signal is inverted meaning our amplifier works as needed, The results we obtained perfectly coincide with the theory.